Reg. No. [redacted] **[18DS7DE41]**

SEVENTH SEMESTER B. Tech EXAMINATION, MARCH–2022

PREDICTIVE ANALYTICS USING PYTHON

Time: 3 Hours **Maximum Marks: 70**

Instructions: i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed.

1. a. Discuss the various applications of predictive analytics in industries which are required to reduce risks, optimize operations and increase revenue? **08**

b. There are two categorical variables such as Gender (Male and Female) and Mathematics test outcome (pass/fail). Find the chi-square value and examine whether there is correspondence between the observed and expected frequency? Justify the reason **06**

	Boys	Girls
Pass	17	20
Fail	8	5

2. a. Differentiate between the univariate and multivariate analysis with the help of plot. Add a note on boxplot. **07**

b. Illustrate and explain with an example about different types of missing data and different types of missing data handling techniques. **07**

3. a. Outline Uniform and Gaussian distribution with a plot and explain them along with their appropriate function. Write the python code to obtain these distributions for random numbers. **10**

b. Given the following pair of variables X and Y: (a) make a scatter diagram., (b) Do you think that there is any correlation between the variables x and y? (c) What are the Merits and demerits of scatter plot **04**

X	Y
2	18
4	12
5	10
6	8
8	7
11	5

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4. a. **Table X** **04**

Index	Marks Scored	Name	Subject_id
1	98	Alex	Sub1
2	90	Amy	Sub 2
3	87	Allen	Sub 4
4	69	Alice	Sub 6
5	78	Ayoung	Sub 5

Table Y

Index	Marks Scored	Name	Subject_id
1	89	Billy	Sub2
2	80	Brain	Sub 4
3	79	Bran	Sub 3
4	97	Bryce	Sub 6
5	88	Bretty	Sub 5

i. Write python code to perform the concatenation of the two dataframe using Table X and Y when axis=0 and write the output.

ii. Write python code to perform the concatenation of the two dataframe using Table X and Y when axis=1 and write the output.

b. The lengths and weights of six samples are given the table. Find the Pearson 's correlation coefficient. Describe the different types of correlation coefficient and add a note on significance of correlation coefficient **10**

Length (X)	Weight (Y)
3	5
8	12
6	11
7	14
10	16
11	17

5. a. Use two independent variable such as age and experience and salary as the dependent variable. Illustrate about the problem caused by applying Linear Regression model and also add a note on various methods to solve multicollinearity problem. **08**

b. Choose the Simple Linear Regression Technique and calculate the intercept, slope and Residual Standard Error by using the following data. Why Residual Standard Error is squared in Simple Linear Regression Technique? **06**

X(Age)	Y(Glucose Level)
43	95
21	69
25	79
42	75
57	87
59	81

OR

6. a. With a neat schematic diagram discuss outlier detection using different statistical methods. Discuss how Local outlier factor overcome the disadvantages associated with the statistical methods. **09**

b. Differentiate between Linear Regression and Logistic regression and describe it with the help of linear and sigmoid curve by considering an appropriate example. **05**

7. a. Compare how Decision Tree and Random Forest Technique diversify each other. Illustrate the Decision Tree and Random Forest Technique with an example using Bagging and Feature Randomness as the important features. **08**

b. Elaborate on the different hyperparameters of Random Forest technique that have significant impact to increase the predictive power of the model. **06**

OR

8. a. Use three different scenarios and explain the scenario to choose the best hyperplane and boundary line in Support Vector Machine **02**

b. Construct the Decision tree using Information gain as the metric by using the table given below which comprises of 14 instances and 4 attributes column and play as 1 target column. **12**

Day	Outlook	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	True	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
5	Rainy	Cool	Normal	False	Yes
6	Rainy	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	cool	Normal	False	Yes
10	Rainy	Mild	Normal	False	Yes
11	Sunny	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rainy	Mild	High	True	No